

# CompTIA A+ Core 1 Playbook

*IT Labs*

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## 2.2.11 Choose & Install a Motherboard

You work at a computer repair store. You are in the process of building a new computer system. The customer wants a motherboard with the following features:

You are in the process of building a new computer system. The customer wants a motherboard with the following features:

- 16x PCI Express slot for video card support
- M.2 storage support
- Support for 64-bit processors
- Dual-channel memory support
- Support for up to 64 GB of memory

In this lab, your task is to:

- Select and install the motherboard that meets the scenario requirements.
- Connect the following computer case cables:
  - Power switch
  - Power LED 3-1 pin
  - HDD LED
  - Speaker
  - Fan front
  - Fan rear
  - USB (Black cable- 2.0)
  - Front panel audio
  - USB 3.0
- Connect the following connectors from the power supply:

- CPU power
- Main motherboard power

## Explanation

Complete this lab as follows:

1. Under Workbench, select **Hardware**.
2. Find and install the correct motherboard.
  1. Above the computer, select the Change View icon and then **Motherboard**.
  2. Under Inventory, expand **Motherboards**.
  3. Find the correct motherboard.
    1. Select a ***motherboard***.
    2. In the Selected Item pane, select the information icon next to the image.
    3. Expand the window for better readability.
    4. Read the *motherboard documentation* in the left pane to identify the supported features.
    5. Close the Item Detail window.
    6. Repeat this process for each motherboard until you find one that meets all the scenario requirements.
  4. For this scenario, drag the **Motherboard, Socket 1151** to the motherboard plate in the computer case.
3. Use the motherboard specifications to locate the required connection ports.
  1. Select the **Motherboard, Socket 1151**, if not already the Selected Item.
  2. In the Selected Item pane, select the information icon next to the image.
  3. Expand the window for better readability.
  4. Use the images and documentation in the left pane to identify the location of the following connectors:
5. The connectors are displayed with the corresponding motherboard diagram location number. Scroll to the bottom of the motherboard details to see the panel diagram to connect items linked to location 13.

| Case Connector                                                 | Location # |
|----------------------------------------------------------------|------------|
| USB 3.0 connector (blue)                                       | 9          |
| Audio Connector                                                | 18         |
| USB 2.0 Connector (black)                                      | 14         |
| 3-pin FAN connector (Fan front)                                | 4          |
| 3-pin FAN connector (Fan rear)                                 | 4          |
| Speaker                                                        | 13         |
| HDD LED (Hard Drive LED)                                       | 13         |
| Power LED 3-1 pin<br><br>(Note: Use the bottom PLED location.) | 13         |
| Power switch (PWRSW)                                           | 13         |

6. Connect the system case connectors.

1. Select the **System Case**.
2. In the Selected Item pane, the connectors are displayed below the case image.
3. Drag each **Connector** to the identified port location on the motherboard. If necessary, zoom in to get a closer view of the motherboard.
4. Repeat this process until all the case connectors are connected to the motherboard.

4. Connect the power supply power cables to the motherboard.

1. Select the **Motherboard, Socket 1151**.
2. In the Selected Item pane, select the information icon.
3. Expand the window for better readability.
4. Identify the location for the following connectors. The location number is included below.

| Power Supply Connectors                       | Location # |
|-----------------------------------------------|------------|
| 8-pin CPU power connector                     | 2          |
| 20+4 Power Connector (Main motherboard power) | 2          |

5. Connect the power supply connectors.
  1. Select the power supply.
  2. Drag the **Connector, Power Supply, 8-pin CPU** connector to the port on the motherboard.
  3. Drag the **Connector, Power Supply, 20+4 pin** connector to the port on the motherboard.